SEM-I

Course code: **22IUCC0101B**	**Course Name:** Mathematics-I
Credits: 4 **Teaching scheme:** L-3 T-1	**Evaluation scheme:** CA–40, ESE–60
Duration of End Sem. Exam: 3 hr.	
Pre-requisites:	
Course Objectives:	
Course Outcomes: At the end of the course, the students will be able to -	
CO1: Understand the basic concepts, principles, rules of algebra to solve the engineering problem in real world.	
CO2: Apply the basic concept of trigonometry to solve the engineering problem.	
CO3: Solve the engineering problem under the given geometrical condition.	
CO4: Utilize the basic concept of statistics to solve engineering problem.	
CO5: Understand the basic concept of system of complex numbers and illustrate simple problems.	
CO6: Understand the basic concept of system of complex numbers and illustrate simple problems.	

Contents:

Unit	Content	Teaching hours
1	**Algebra** Logarithm: Concept and Definition of Logarithm, Rules of Logarithms (Simple examples only),Permutation and Combination: Introduction, Fundamental Principle of counting, Permutations, Combinations. (Simple examples only), Binomial Theorem: Introduction, Binomial theorem for positive integral indices, General and middle terms. (Simple examples only) Partial Fraction: Introduction, Definition and types of fraction, Resolve into partial based on nature of factors and related problems	8
2	**Determinant and Matrices** Determinant :Definition and simple examples of determinant of order 2 & 3, Properties of Determinant, Minor and Cofactors, Crammer's Rule ,Matrices: Concept and definition of matrices Types of matrices, Algebra of matrices, Transpose, orthogonal, Adjoint of matrices, Inverse of matrix, Equivalent matrices, Matrix inversion method	8
3	**Trigonometry** Trigonometric ratios of compound, allied, multiple and sub multiple angles, Factorization and De- Factorization formulae, Inverse trigonometric ratios and related problems, Principle values and relation between trigonometric and inverse trigonometric ratios. Properties of triangle: sin, cosine, projection and tangent rule.(Without proof)	10
4	**Co-ordinate Geometry** Distance Formula, Section Formula, Different centers of triangle (without proof), Shifting of origin, Slope of line, Forms of Straight lines, Point of intersection of two lines, Angle between two lines (without proof) and Condition of parallel and perpendicular lines, Perpendicular Distance of a point from the line, Perpendicular Distance between two parallel lines. Conic Section: Circle, Parabola, Ellipse, Hyperbola.	10
5	**Data Analysis** Measure of central tendency: Mean (AM, GM, HM,) Mode, Median. , Measure of dispersion: Range, MD, SD, and Test of consistency	6
6	**Complex Numbers** Definition of complex number, Algebra of Complex number, Argand diagram, Forms of Complex number, Properties of modulus and arguments, De Moivre's Theorem, Cube root of unity.	6

Reference Books:

1.Hall and Knight, Higher Algebra Arihant Publications India limited,
2. Loney, SL (1912), Plane Trigonometry, Cambridge: University Press, JFM 25.0892.12
3. Loney, SL (1895), The elements of coordinate geometry, London: Macmillan and Co.
4. S.C. Gupta & V.K. Kapoor, Fundamentals of Mathematical Statistics, 11th edition, Sultan Chand & Sons, New Delhi.
5. David Freedman, Robert Pisani and Roger Purves, Statistics, 4th Edition.
6. S.C. Gupta, Fundamental of Statistics, Himalaya.
7. H.C. Saxena, Mathematical Statistics, Sultan Chand & Sons, New Delhi.

Online Resources:

1. www.scilab.org/-SCI Lab
2. www.mathworks.com/products/matlab/ -MATLAB
3. www.allmathcad.com/- MathCAD
4. www.wolfram.com/mathematica/ - Mathematica

Course code: 22IUCC0102B	**Course name: Physics-I**	**Course category:** PCC
Credits: 4 **Teaching scheme:** L-3 T-1		**Evaluation scheme:** CA–40, ESE–60
Pre-requisites:		
Course Objectives:		
1.		
Course Outcomes: At the end of the course, the students will be able to -		
CO1: dentify different systems of units and convert units from one system to another as well as conversant with Practical units.		
CO2:Understand equations of motion and their applications.		
CO3:Understand equations of motion and their applications.		
CO4:Acquire basic knowledge on semiconductor & application of P-N junction diode		
CO5:Use the properties of photoelectric effect, optical fiber &nano particle for various engineering application		
CO6: Understand concept of LIGHT		

Contents:

Unit	Content	Teaching hours
1	**Units & Measurements** Introduction of Unit, fundamental and derived quantities and their units, different system of Units (CGS, MKS, FPS and SI), Definition of accuracy and error, Estimation of Errors, absolute error, relative error and percentage error. Numerical problems. Explanation of Dimension of physical quantity Dimensional Equations of physical quantities and their uses with examples.	8
2	**Mechanics** Introduction to scalar and vector Quantities, representation of vector, addition, subtraction and multiplication of vectors, parallelogram law of vector Newton's Laws of motion: First law, Second law, &Third law, Statements & examples, Principle of conservation of linear momentum, statement and simple examples. Work, power and energy, its concept, units and dimension, Potential and Kinetic energy, its mathematical relations, Principle of conservation of energy, Recoil of gun.	8

3	**Circular Motion** Circular motion, and angular velocity, relation between angular velocity and linear velocity, Rotational motion, Simple Harmonic Motion, derivation of its equationsdefinition of amplitude, time period, frequency, phase etc., mathematical relations and units, Simple Pendulum	**6**
4	**Heat And Thermodynamics** Concept of heat and temperature, Unit of heat, Joule and calorie, thermometer, its interval and fixed point, different scales of temperatures and their conversion formulae. Transmission of heat, three modes of heat transfer, conduction, convection and Radiation, good and bad conductor of heat, coefficient of thermal conductivity, its S.I. unit and dimension. 1st law and 2nd law of thermodynamics, Joule's law & its derivation.	**8**
5	**Modern Physics** Photo electricity, Photon (quantum), Plank's hypothesis, energy of photon, properties of photons. Photo electric effect: Circuit diagram, process of photoelectric emission, definitions:- threshold frequency, threshold wavelength, Stopping potential, characteristics of photoelectric effect, Application of photoelectric cell & LDR.	**6**

Text Books:
Reference Books: 1. Engineering Physics- R K Gour& S L Gupta, DhanpatRai Publication- New Delh 2.Engineering Physics- M N Avadhanulu& P G Kshirsagar, S Chand & Company Ltd.- New Delhi 3.Fundamental of Physics- Jearl Walker, Wiley India Edition 4.Engineering Physics- V Rajendran, Tata McGraw-Hill Publishing Company Ltd- New Delhi 5.Applied Physics- Manpreet Singh, Dr. Major Singh, Mrs.Hitashi Gupta, S K Kataria& Sons- New Delhi
Online Resources 1. http://nptel.ac.in/course.php?disciplineId=115 2.http://nptel.ac.in/course.php?disciplineId=104 3.http://hperphysics.phy-astr.gsu.edu/hbase/hph.html 4.www.physicsclassroom.com 5.www.physics.org 6.www.fearofphysics.com

Course code: 22IUCC0103B	**Course name:**Chemistry-I
Credits: 4 **Teaching scheme:** L-3, T-1	**Evaluation scheme:** CA–40, ESE–60
Duration of End Sem Exam: 3hr	
Pre-requisites:	
Course Objectives:	
Course Outcomes: At the end of the course, the students will be able to -	
CO1: Demonstrate an understanding of basic principles of organic chemistry and how they relate to everyday Experiences.	
CO2: Illustrate the Basic Concept of Structure of Atom	
CO3: Describe nature of bonding in molecular compounds.	
CO4: To know the common analysis techniques used in metallurgical industries.	
CO5: Students know how to approach the problem of choosing an alloy for a particular application.	
CO6: Use Relevant Electrolyte in Electrolysis for different Applications	

Contents:

Unit	Content	Teaching hours
1	**Basic principles of organic chemistry** Introduction, Structural representation of organic molecules-Condensed formula, structural formula, Bond line or Zigzag formula with examples. Classification of organic compound. Acyclic/Open chain compound. Cyclic compound , Homocyclic,Heterocycliccompound,Alicyclic,Aromatic(Benzenoid,Nonbenzenoid),Heterocyclic (Aromatic . Nonaromatic),(Definition with example.) ,Functional group in organic compounds. Halides,Cyanides,Isocyanides, Nitro compound,Alcohol,Phenol,Amines.Ether,Aldehydes,Ketones,Carboxylic acid, Ester, Amide, Sulphonic acid, Acyclic chloride,Acidunhydride,Alkyne,Alkene,Alkane.(Structure and Functional group with examples).	8
2	**Atomic Structure and Chemical bonding** Introduction ,Definition of atom, Bohr's atomic model, definition of atomic number, atomic mass number and their differences, , Isotopes and Isobars: Definitions, examples and distinction , Orbits and orbitals: s, p, d, f orbitals, shapes and description of s-orbital and p-orbital. Distribution of electrons in orbitals: , Definition of electronic configuration, Aufbau's principle, Hund's rule, orbital Electronic configurations (s, p, d, f) of elements having atomic number 1 to 30	6
3	**Chemical bonding** Introduction, Valance Definitions of valence. Definition of electrovalence, positive and negative electrovalence, formation of Electrovalent compounds-MgO,CaCl2 , Valance Definitions of valence. Definition of electrovalence, positive and negative electrovalence, formation of Electrovalent compounds-MgO,CaCl2 , Definition of covalence, single, double and triple covalent bonds, formation of Covalent compounds H2O,CO2,N2	6
4	**Metals and Ores**	8

	Introduction , Occurrence of metals in free and combined state, definitions- mineral, ore, Gangue, flux and slag, metallurgy. , Ores of Iron , Steps involved in the extraction of metals: Crushing, Concentration, Reduction, Refining , Mechanical properties of metals- Hardness, ductility, malleability, tensile strength, toughness, mach inability, weld- ability, forging, soldering, brazing, cast ability.	
5	**Alloys and its applications** 5.1 Introduction , Definition of alloy, purposes of making alloys with examples. Preparation methods- Fusion, Compression , Classification of Alloys- Ferrous and nonferrous alloys with examples. , Examples of alloys-Composition, properties and application of duralumin, Woods metal. Brass, Solders. ,Faraday's laws of electrolysis: Faraday's first and second law, relation between electrochemical equivalent and chemical equivalent. ,. pH and pOH: Definition of pH, pOH, pH Scale	6
6	**Electrochemistry** Introduction , Basic concepts of electrolysis: Electrolyte, types of electrolyte- strong and weak electrolyte, their difference. , affecting degree of ionization., Applications of electrolysis- Electroplating of silver, electro refining of blister copper, , Faraday's laws of electrolysis: Faraday's first and second law, relation between electrochemical equivalent and chemical equivalent. , pH and pOH: Definition of pH, pOH, pH Scale	6

Text Books:

1. Barrow G.M., Physical Chemistry, McGraw-Hill Publication, New Delhi.
2. ShikhaAgrawal, Engineering Chemistry- Fundamentals and Applications, Cambridge Publishers.
3. Atkins, Physical Chemistry, OUP Oxford.
4. Engineering Chemistry, Wiley India Pvt, Ltd, First edition

Reference Books:

1. Jain P.C & Jain Monica, Engineering Chemistry, DhanpatRai& Sons, Delhi.
2. Bhal&Tuli, Text book of Physical Chemistry, S. Chand & Company, New Delhi.
3. O. G. Palanna , Engineering Chemistry, Tata McGraw-Hill Publication, New D
4. S. S. Dara, A textbook of Engineering Chemistry, McGraw-Hill Publication, New Delhi.
5. ShashiChawla, Engineering Chemistry, DhanpatRai and Co Education and Technical Publishers
6. V. R Gowarikar, Polymer Science, New Age International Ltd.
7. B. K. Sharma, Instrumental Methods of Chemical Analysis, Krishna Prakashan Media

Online Resources:

1. NPTEL Basic Courses Engineering Chemistry (https://nptel.ac.in/courses/122/101/122101001/)
2. https://www.ncertbooks.guru/engineering-chemistry/
3. Coursera Chemistry Courses (https://www.coursera.org/browse/physical-science-and-engineering/chemistry?languages=en&page=2)

Course Code: 22IUCC0104H	**Course Name:**English-I
Credits:3 **Teaching Scheme:** L-3	**Evaluation Scheme:** CA–40, ESE–60
Duration of EndSemExam: 3hr	
Pre-requisites:	
Course Objectives:	
Course Outcomes: At the end of the course, the students will be able to -	
CO1: Student will be able to apply basic concept of grammar in day to day communication	
CO2: Student will be able to Summaries and comprehend the given passages	
CO3: Student will be able to compose dialogue and paragraph for different situations	
CO4: Student will be able to use appropriate words as per the context	
CO5: Student will be able to deliver prepared speeches to express ideas, emotions and thoughts.	

Contents –

Unit	Content	Teaching hours
1	**Basic Grammar** 1.1 Articles, 1.2 Preposition, 1.3 Conjunction, 1.4 Tenses, 1.5 Direct indirect speech, 1.6 Change the voice.	10
2	**Comprehension Skills** 2.1 Definition of Reading 2.2 Definition of Reading Comprehension 2.3 Kinds of Reading 2.4 Extensive Reading 2.5 Intensive Reading 2.6 Effective Strategies for Reading Comprehension	10
3	**Paragraph and dialogue writing** 3.1 Paragraph Writing: Elaborate and expand the ideas with cohesion, coherence and use of correct punctuation marks 3.2 Types of Paragraph: Narrative, Descriptive, Technical, Comparison and Contrast 3.3 Dialogue Writing: Based on various situations	10
4	**Vocabulary building** 4.1 Words Often Confused 4.2 Collocation 4.3 Prefix and Suffix 4.4 Synonyms and Antonyms	6
5	**Speeches** Speech Writing based on situations	6

	5.1 Welcome Speech, 5.2 Farewell Speech, 5.3Vote of Thanks and Introducing a Guest	
6	**Introduction to the sounds of English** 6.1 English Phonetics 6.2 Phonetics symbols 6.3 Consonant and vowel sounds 6.4 Accent, rhythm and intonation. 6.5 The functions of intonation	6

Reference Books:

1. English Grammar and Composition Book (Wren & Martin)
2. Oxford Modern English Grammar
3. Fundamentals of English Grammar (N.C.Sinha)

Online Resources:

1. https://www.britishcouncil.in/english/learn-online
2. http://learnenglish.britishcouncil.org/en/content
3. http://www.talkenglish.com
4. Languagelabsystem.com
5. www.wordsworthelt.com

Course Code: 22IUCC0105T		**Course Name:** Fundamental of Computer System
Credits: 2	Teaching **scheme:** L-2	**Evaluation Scheme:** Audit
Duration of End sem Exam: --		
Pre-requisites:		
Course Outcomes: At the end of the course, the students will be able to -		
CO1: Use Computer System and its Peripherals**.**		
CO2: Prepare Business Document using word processing tool.		
CO3: Interpret data and represent it graphically using spreadsheet.		
CO4: Prepare Professional Presentations.		
CO5: Identify the layout of wired and wireless LAN Environment		
CO6: Prepare and interpret database using csv file.		

Contents:

Unit	Content	Teaching hours
1	**Introduction to Computer System** Basic of computer system: Overview of Hardware and Software, block diagram of Computer System, Input/output unit CPU, Arithmetic logic Unit (ALU), System of software Memory Unit. 1.2 Generation of O.S.: Overview of Different Types and Generation of Operating system 1.3 Working with OS: Create and manage file and folders, Copy a file, renaming and deleting of files and folders, Searching files and folders, application installation, creating shortcut of application on the desktop. 1.4 Printer and Scanner: Overview of printer and scanner, Features of printer and Scanner, Difference between Printer and Scanner.	4
2	**Word Processing** Word processing: Overview of Word processor Basics of Font type, size, color, Effects like Bold, italic, underline, Subscript, superscript, Case changing options Previewing a document, Saving a document, Closing a document and exiting application. 2.2 Editing a document: Format and align text, Formatting Paragraphs, Line and paragraph spacing, using FIND and REPLACE, Setting line spacing, add bullet and numbers in lists, add borders and shading, document views, Page settings and margins, Spelling and Grammatical checks. 2.3 Inserting an image to word document: Insert and delete a page break, Insert page numbers, Insert the date and time, Insert special characters (symbols), Insert a picture from a file, Resize and reposition a picture 2.4 Working with table: Insert a table, Convert a table to text, Navigate and select text in a table, Resize table cells, Align text in a table, Format a table, Insert and delete columns and rows, Borders and shading, Repeat table headings on subsequent pages, Merge and split cells.	4
3	**Spreadsheet** Working with spread sheet: Overview of workbook and worksheet, Create Worksheet Save, Copy Worksheet, Delete Worksheet, Close and open Workbook 3.2 Editing worksheet: Insert and select data, adjust row height and column width, delete, move data, insert rows and columns, Copy and Paste, Find and Replace, Spell Check Zoom In-Out, Special Symbols, Insert Comments, Add Text Box, Undo Changes, Freeze Panes, hiding/unhanding rows and columns. 3.3 Formatting cells and sheet: Setting Cell Type, Setting Fonts, Text options, Rotate Cells,	4

	Setting Colors, Text Alignments, Merge and Wrap, apply Borders and Shades, Sheet Options, Adjust Margins, Page Orientation, Header and Footer, Insert Page Breaks, Set Background. 3.4 Working with formula: Setting Cell Type, Setting Fonts, Text options, Rotate Cells, Setting Colors, Text Alignments, Merge and Wrap, apply Borders and Shades, Sheet Options, Adjust Margins, Page Orientation, Header and Footer, Insert Page Breaks, Set Background 3.5 Advanced operation: Conditional Formatting, Data Filtering, Data Sorting, Using Ranges, Data Validation, Adding Graphics, Printing Worksheets, print area, margins, header, footer and other page setup options 3.6 Working with chart: Introduction to charts, overview of different types of charts, Bar, Pie, Line charts, creating and editing charts. Using chart options: chart title, axis title, legend, data labels, Axes, grid lines, moving chart in a separate sheet.	
4	**Presentation tool** Creating presentation : Outline of an effective presentation, Starting a Presentation Files, Creating a Basic Presentation, Working with textboxes, Apply Character Formats, Format Paragraphs View a Presentation, Saving work, creating new Slides, Applying a theme, Changing Colors, fonts and effects, apply custom Color and font theme, changing the background, Arrange Slide sequence. 4.2 Working with table: Insert a Table in a Slide, Format Tables, and Import Tables from Other Office Applications. 4.3 Animation: Adding and Modifying animation to a Presentation - Insert Images into a Presentation, insert audio clips, video/animation, Add Shapes, Add Visual Styles to Text in a Presentation, Edit Graphical Objects on a Slide, Format Graphical Objects on a Slide, Group Graphical Objects on a Slide, Apply an Animation Effect to a Graphical Object, Add Transitions, Add Speaker Notes, Print a Presentation. 4.4 Working with Charts: Insert Charts in a Slide, Modify a Chart, Import Charts from Other Office Applications.	4
5	**Basic of Networking** Introduction of networking Devices: Application of Network interface card (NIC), HUB, Switches, Routers, Modem 5.2 Concept of Network :- LAN,MAN,WAN Wireless network and devices; Wi-Fi, Access point, Repeaters, Bluetooth 5.3 Characteristics of Cable: CAT5,CAT6,fibre optic cable, use of crossover and straight cable,RJ-45 connectors, SC.ST,FC,LC type fiber Connectors 5.4 Types of Servers	4
6	**Databases** Create an appropriate database structure : Import data from existing files (including .csv, .txt) using specified field names to create tables. Create relationships between tables Create and use a data entry form . Types of database Characteristics, uses, advantages and disadvantages of a flat file and a relational database .Primary and foreign keys Characteristics of primary key and foreign keys. Form design Characteristics of good form design.Perform calculations Use arithmetic operations or numeric functions to perform calculations including calculated fields, calculated controls Use formulae and functions to perform calculations at run time including addition, subtraction, multiplication, division, sum, average, maximum, minimum, count .Sort data Use a single criterion, or multiple criteria to sort data into ascending or descending order.Search and select data Use a single criterion, or multiple criteria to select subsets of data using a query perform searches using a variety of operators including AND, OR, NOT, LIKE, >, =, <=, <> perform searches using wildcards.Display data Produce reports to display data including displaying all the required data and labels.	6

Text Books:
1. B Ram –Computer Fundamental architecture and Organization, New age international Publications 2. Dr RajendraKawale ,"Computer Fundamental", Devraj Publications Dist. Solapur

Course Code: 22IUCC0106T	**Course Name:** Engineering Graphics
Credits:3 **Teaching Scheme:** L-3	**Evaluation Scheme:** CA–40, ESE–60
Duration of EndSemExam: 3hr	
Pre-requisites:	
Course Objectives:	
Course Outcomes: At the end of the course, the students will be able to -	
CO1: Student will able to draw geometrical figures with proper dimensioning technique	
CO2: Student will able to draw the views of given object by using principles of orthographic projection	
CO3: Student will able to draw isometric views of given component or from orthographic projection	
CO4: Student will able to draw Engineering curves	
CO5: Student will able to draw free hand sketches of given engineering Elements.	
CO6: Student will able to create 2D computer drawings on CAD software	

Contents –

Unit	Content	Teaching hours
1	**Basics of Graphics:** Introduction to Engineering Graphics, Need of Engineering Drawing, Drawing Instruments, BIS code of practice for general engineering drawing, standard sizes of drawing sheets (ISO-A Series) , single stroke vertical Letters and Numbers, convention of lines and their applications, Scale- reduced, enlarged and full size, Dimensioning technique, and Geometrical construction	4
2	**Orthographic Projection:** Orthographic projection-first angle and third angle method their symbol, conversion of pictorial view into orthographic views- Orthographic projection of machine parts containing plain surfaces, slanting surfaces slots, ribs, cylindrical surfaces (First angle method is to be used)	7
3	**Isometric Projections:** Introduction to Pictorial views, Isometric scale and natural scale, Converting Orthographic Projections into Isometric Projections and Isometric views. Illustrative problems related to simple objects having plain, slanting , cylindrical surfaces and slots on slanting surfaces	7
4	**Engineering Curves** Curves used in Engineering Practice, Conic sections, Construction of conics by different methods, Rectangular- hyperbola, Cycloidal curves, Epi and hypo-cycloids, Involutes and Archimedean spiral, Loci of points on single slider crank mechanism with given specification	7
5	**Free Hand Sketches of Engineering objects** Free hand sketches of machine elements; Thread profiles, nuts bolts, studs set screws, washers, locking arrangments.	3
	Introduction to CAD	

6	Listing the computer technologies that impact on graphical communication, Demonstrating knowledge of the theory of CAD software [such as: The Menu System, Toolbars (Standard, Object Properties, Draw, Modify and Dimension), Drawing Area (Background, Crosshairs, Coordinate System), Dialog boxes and windows, Shortcut menus (Button Bars), The Command Line (where applicable), The Status Bar, Different methods of zoom as used in CAD, Select and erase objects	4

Reference Books:

1. N. D. Bhatt and Panchal V. M., "Engineering Drawing", Charotar Publishing House, Anand
2. P. J. Shah , " A Text Book of Engineering Drawing", S.Chand, New Delhi
3. Shah M.B. &RanaB.C,"Engineering Drawing & Computer Graphics", Pearson Publications,
4. .Agrawal B.&Agrawal C.M, "Engineering Graphics", TMH Publications
5. Narayana K.L. & P. Kannaiah, "Engineering Drawing", Scitech Publications
6. .P.I Vargese, "Engineering Graphics", McgrawHill Publications
7. D.A.Hindoliya," Engineering Graphics", B.S.Publications
8. Dabhade M. L., "Engineering Graphics", Vol.-I and Vol.-II, Vision Publications, Pune .
9. K.Venugopal, "Engineering Drawing and Graphics" ,New Age International Publishers

Online Resources:

1. https://www.machinedesignonline.com
2. https://www.me.umn.pdu/courses/me2018/handouts/engg%20graphics.pdf
3. https://www.youtube.com/watch?v=fvjk7PlxAuo
4. https://www.youtube.com/watch?v=3WXPanCq9LI

Course code: 22IUCC0102L	**Course name:** Physics-I Lab
Credits: 1 **Teaching scheme:** P-2	**Evaluation scheme:** TW–50, PR–0
Pre-requisites:	
Course Outcomes: At the end of the course, the students will be able to -	
CO1. Identify different systems of units and convert units from one system to another as well as conversant with Practical units.	
CO2. Understand equations of motion and their applications.	
CO3: Understand equations of motion and their applications.	
CO4. Acquire basic knowledge on semiconductor &applicatation of P-N Junction diode.	
CO5. Use the properties of photoelectric effect, optical fiber &nano particle for various engineering application	
CO6: Understand concept of LIGHT	

Contents –
Perform any 10 practicals from the list.

Practical	Content	Teaching hours
1	Measure the dimensions of given objects using Vernier caliper.	2
2	Measure the dimensions of given objects using micrometer screw gauge	2
3	Measure the dimensions of given objects using spherometer	2
4	To determine the valve of acceleration due to gravity [g] of a place with simple pendulum.	2
5	Determination of force constant using Helical Spring	2
6	Use Searle's thermal conductivity apparatus to find coefficient of thermal conductivity of a given material	2
7	Use Joule's calorimeter to determine Joule's Mechanical/ electrical equivalent of heat	2
8	To study the I-V characteristics of photoelectric effect	2
9	To determine the refractive index of the material of the glass slab by pin method.	2

Course code:22IUCC0103L	**Course name:**Chemistry-I Lab
Credits: 1 **Teaching scheme:** P-2	**Evaluation scheme:** TW–50, PR–0
Pre-requisites:	
Course Objectives:	
Course Outcomes: At the end of the course, the students will be able to	
CO1: Demonstrate an understanding of basic principles of organic chemistry and how they relate to everyday Experiences. **CO2:** Illustrate the Basic Concept of Structure of Atom **CO3:** Describe nature of bonding in molecular compounds. **CO4:** To know the common analysis techniques used in metallurgical industries. **CO5:** Students know how to approach the problem of choosing an alloy for a particular application. **CO6:** Use Relevant Electrolyte in Electrolysis for different Application	

Contents –

Unit	Content	Teaching hours
1	Do you know chemistry laboratory	2
2	Determine preliminary test and functional group of organic compound No.1	2
3	Determine preliminary test and functional group of organic compound No.2	2
4	Determine preliminary test and functional group of organic compound No.3	2
5	Determine preliminary test and functional group of organic compound No.4	2
6	Identify anion in the given ionic solution (non- metallic ions)	2
7	Identify cation in the given ionic solution (metallic ion)	2
8	Determine the percentage of iron in the given sample using redox titration	2
9	Determine the pH value of given solution using universal indicator	2
10	Determine the voltage generated from chemical reaction using Daniel cell.	2
11	Determine the electrode potential of Iron metal.	2
12	Determine the rate of corrosion on different temperatures for Aluminum	2

Course code: 22IUCC0104L	**Course name:** Language Lab-I
Credits: 1**Teaching scheme:** P-02	**Evaluation scheme:** TW–50, PR–0
Pre-requisites:	
Course Outcomes: At the end of the course, the students will be able to –	
CO1: Student will be able to apply basic concept of grammar in day to day communication	
CO2: Student will be able to Summaries and comprehend the given passages.	
CO3: Student will be able to Compose dialogue and paragraph for different situations.	
CO4: Student will be able to use appropriate words as per the context	
CO5: Student will be able to Deliver prepared speeches to express ideas, emotions and thoughts	

Contents:

Unit	Content	Teaching hours
1	Make sentences using correct articles	2
2	Make sentences using correct Preposition	2
3	Make sentences using correct Conjunctions	2
4	Make sentences using correct active and passive voice.	2
5	Make sentences using correct direct and in direct speech.	2
6	Make sentences using correct tenses	2
7	Make oral presentations using correct grammar.	2
8	Write short paragraphs emphasizing on syntax and sentence structure.	2
9	Write different types of dialogues for role plays.	2
10	Construct sentences using various collocations.	2
11	Write different types of speeches using new vocabulary.	2
12	Deliver extempore short speeches of 3to5 minutes.	2

Course Code: 22IUCC0105L	**Course Name:** Fundamental of Computer System Lab
Credits: 1 Teaching **Scheme:** P-2	**Evaluation Scheme:** TW–50, PR–0
Duration of End Sem Exam:	
Pre-requisites:	
Course Outcomes: At the end of the course, the students will be able to –	
CO1: Use Computer System and its Peripherals**.**	
CO2: Prepare Business Document using word processing tool.	
CO3: Interpret data and represent it graphically using spreadsheet.	
CO4: Prepare Professional Presentations.	
CO5: Identify the layout of wired and wireless LAN Environment	
CO6: Prepare and interpret database using csv file.	

Contents –

Unit	Content	Teaching hours
1	Identify various Input/output devices. Connections and peripherals of computer system.	2
2	Install and work with wifi printer and scanner setting.	2
3	Create, edit and save documents: apply formatting features on the text, line, paragraph. Use bullets, numbering, page formatting.	2
4	Insert, edit images and shapes, sizing, cropping color, background, group/ungroup. Insert and apply various table formatting features on it.	2
5	Create, open and edit worksheet 1.Enter data and format it adjust row height and column width. 2. Insert and delete cells, rows and columns. 3. Apply wrap text, orientation feature on cell.	2
6	Insert formulas, ”IF” conditions, functions and named ranges in worksheet. Apply data sort, Filter and data validation features.	2
7	Create charts to apply various chart options.	2
8	Create slide presentation. 1. Apply design themes to given presentation. 2.Add new slides and insert pictures / images, shapes. 3. Add tables and charts in slides. 4.Run slides presentation in different modes. 5. Print slide presentations on handouts.	2
9	Apply animation effects to text and slides Add audio and video files in presentation.	2

10	Connect/Disconnect LAN cable, external hard disk modem. Connect desktop computers and laptops with LCD/DLP projectors.	2
11	Prepare and test crossover and straight cable, CAT5,CAT6 cable, using crimping Tools, splicer. Connect two switches/Hubs/switches/r outers physically.	2
12	Identify server by its type and verify its Specification	2
13	Create new database, table and build relationship between them, saving tables	2
14	Managing data in tables and performing calculation using functions, and sorting data from tables.	2

Course Code: 22IUCC0106L	**Course Name:** Engineering Graphics Lab
Credits:2 **Teaching Scheme:** P-4	**Evaluation Scheme:** TW–100, PR–0
Duration of End SEM Exam:	
Pre-requisites:	
Course Outcomes: At the end of the course, the students will be able to –	
CO1: Student will able to draw geometrical figures with proper dimensioning technique	
CO2: Student will able to draw the views of given object by using principles of orthographic projection	
CO3: Student will able to draw isometric views of given component or from orthographic projection **CO4:** Student will able to draw Engineering curves	
CO5: Student will able to draw free hand sketches of given engineering Elements	
CO6: Student will able to create 2D computer drawings on CAD software	

Contents –

Unit	Content	Teaching hours
1	Draw different types of lines, dimensioning styles. Draw regular geometric construction	2
2	Draw one figure showing dimensioning techniques. Draw one problem on redraw the figure.	2
3	Draw two problem on orthographic projection using first angle method of projection having plain and slanting surface	2
4	Draw two problem on orthographic projection using first angle method of projection having cylindrical surface, rib	2
5	Draw two problems on Isometric view of simple object having plain and slanting surface using natural scale	2
6	Draw two problems on Isometric view of simple object having plain and slanting surface using natural scale	2
7	Draw two problem on Isometric projection of object having cylindrical surface by using isometric scale	2
8	Draw two problem on isometric projection of object having slot on slanting surface by using isometric scale	2
9	Draw Engineering curves generated from conics 1. Ellipse 2. Parabola 3. Hyperbola	2
10	Draw Involutes of circle and pentagon. Draw cycloid curve	2

11	Draw free hand sketches/conventional representation of machine elements such as thread profile, nuts, bolts, studs, set screws, washer, locking arrangement	2
12	Draw free hand sketches/conventional representation of machine elements such as thread profile, nuts, bolts, studs, set screws, washer, locking arrangement	2
13	Draw basic 2D entities like: rectangle, polygon, circle, Arc, circular and rectangular array by using AutoCAD	2
14	Draw basic branch specific components in 2D using AutoCAD	2
15	Draw complex branch specific components in 2D using Autocad	2

SEM-II

Course Code: 22IUCC0201B	**Course Name:** Mathematics-II
Credits: 4 **Teaching scheme:** L-3 T-1 P-0	**Evaluation Scheme:** CA–40, ESE–60
Duration of End Sem Exam: 3 hr	
Pre-requisites:	
Course Outcomes: At the end of the course, the students will be able to -	
CO1: Understand the concept to formulate a mathematical model using of function, and concept of Limits and continuity which rise to derivative.	
CO2: Estimate, to graph derivatives of the function and used rules of differentiation to solve problem involving rates of change, approximation of function and optimization problem.	
CO3: Understand the concept of integration to find the arc length, area, volume and surface area using integrals.	
CO4: Solve ordinary differential equations and also apply these methods to solve engineering problems.	
CO5: Trace the Cartesian curve, parametric curve and Polar curve	

Contents –

Unit	Content	Teaching hours
1	**Function limit and Continuity** Definition of Function, Ways to represent a function, Types of functions, Mathematical modeling, and transformation of functions. Limit: Concept and definition of limit using tangent and velocity problem, Laws of limits with simple problems. Concept of continuity	8
2	**Differential Calculus** Concept and definition of Derivatives through physical and geometrical problem. And Rates of Change, The Derivative as a Function. Rules of Differentiation, Derivative of inverse functions, Implicit Differentiation, Logarithmic Differentiation, Parametric Differentiation, Second order differentiation	8
3	**Application of Derivative** Rate of Change in the Natural and Social Sciences, Exponential Growth and Decay, Linear Approximations and Differentials, Maximum and Minimum Values, Indeterminate Forms and L'Hospitals Rule. Optimizing problems**.**	8
4	**Integral Calculus** Indefinite and Definite integrals. Methods and Properties, Applications to find arc length, Area, Volume and surface area.	10
5	**Differential Equation** Introduction to Initial Value problems. Linear and Bernoulli's equations, Exact equations	8

	and reducible to exact form. Applications of first order differential equations- Orthogonal trajectories, growth and decay problems, Electrical circuits, and falling bodies.	
6	**Curve Tracing** Tracing of standard curves in Cartesian form, Parametric form and Polar form	6

Reference Material:
1. Narayan, Shanti and Mittal, P.K. Differential Calculus, S Chand publication. 2. Narayan, Shanti and Mittal, P.K. Integral Calculus, S Chand publication 3. S.C. Gupta & V.K. Kapoor, Fundamentals of Mathematical Statistics, 11th edition, Sultan Chand & Sons, New Delhi. 4. Hughues- Hallett Gleason, Calculus Single and Multivariable, 4th edition, Wiley. 5. Erwin Kreyszig, Advanced Engineering Mathematics, John Wiley & Sons, New York. 6. B. S. Grewal, Higher Engineering Mathematics, Khanna Publishers, New Delhi. 7. P. N. Wartikar, J. N. Wartikar, Applied Mathematics (Vol I & II), Pune VidyarthiGrihaPrakashan, Pune. 8. H. K. Das and RajnishVerma, Higher Engineering Mathematics, S. Chand & CO. Pvt. Ltd., New Delhi.
Online Resources: 1.www.scilab.org/-SCI Lab 2.www.mathworks.com/products/matlab/ -MATLAB 3.www.allmathcad.com/-MathCAD 4.www.wolfram.com/mathematica/ - Mathematica 5.www.dplot.com/-DPlot

Course Code: 22IUCC0202B	**Course Name:**Physics-II
Credits: 4 **Teaching Scheme:** L-3 T-1	**Evaluation Scheme:** CA–40, ESE–60
Pre-requisites: Soil	
Course Outcomes: At the end of the course, the students will be able to -	
CO1: To learn about the useful solid material in industry & understand its physical properties.	
CO2: To learn about the useful liquid material & understand its physical properties.	
CO3: Understand the concepts of sound wave	
CO4: To learn about the concepts of radioactive and ultrasonic principles.	
CO5: To learn about the concepts of electrostatic & X rays.	
CO6: To learn about the concepts of the principles of electricity , magnetism & semiconductors to solve engineering problems.	

Contents –

Unit	Content	Teaching hours
1	**Properties of Solid** 1.1 Deforming Force and Restoring Force, Elasticity, Plasticity, Rigidity 1.2 Stress and Strain and their types, Elastic limit and Hooke's law Stress -Strain diagram, Poisson's ratio, factors affecting elasticity, With numerical problem	6
2	**Properties of Liquid** 2.1 Fluid friction, pressure, pressure- depth relation, Pascal's law, Archimedes' principle 2.2 Viscosity, velocity gradient, Newton's law of viscosity. 2.3 Free fall of spherical body through viscous medium and Stokes' law, derivation of coefficient of viscosity 'ç' by Stokes' method, effect of temperature and adulteration on viscosity of liquids With numerical problem	8
3	**Wave Theory & Sound** 3.1 Wave , Longitudinal Wave, Transverse Wave, , wave length, amplitude, frequency, time-period and velocity of wave, relation between velocity, frequency and time - period of wave. 3.2 Sound waves, Newton's formula for velocity of sound, Musical sound & noise, Loudness, Pitch, Quality of sound (Timbre), Musical Scale.	8
4	**Radioactivity** 4.1 Radioactivity, α, β and γ rays and their properties. 4.2 Radioactive decay law, half-life period. Energy Mass Relation, Nuclear fission, Nuclear fusion. 4.3 Ultrasonic waves, properties of Ultrasonic waves, Applications of ultrasonic waves, Doppler Effect and its applications.	6
	Electrostatics & X Ray	

5	5.1 Capacitors and capacitance, Parallel plate capacitor, effect of dielectric on capacitance, Series Combination of capacitors, Parallel Combination of capacitors, energy stored in a capacitor. Cells, emf of cell, internal resistance of cell 5.2 X ray Definition, Production of X-rays by modern Coolidge tube, Continuous & Characteristic X ray, Properties of X ray and Engineering , Medical, & Scientific Applications of X ray.	6
6	**Electricity , Magnetism & Semiconductor** 6.1Coulomb's law, Ampere's law Faraday's law motion of charges in electric & magnetic fields to determine e/m. Electrostatic focusing system. Magneto static focusing system. 6.1 Conductors, Insulators And Semiconductors, Energy bands, intrinsic and extrinsic semiconductors p-n junction diode, I-V characteristics of p-n junction, applications of p-n junction diode.	6

Reference Material :

1. Engineering Physics- R K Gour& S L Gupta, DhanpatRai Publication- New Delh

2. Engineering Physics- M N Avadhanulu& P G Kshirsagar, S Chand & Company Ltd.- New Delhi 3. Fundamental of Physics- Jearl Walker, Wiley India Edition

4. Engineering Physics- V Rajendran, Tata McGraw-Hill Publishing Company Ltd- New Delhi

5. Applied Physics- Manpreet Singh, Dr. Major Singh, Mrs.Hitashi Gupta, S K Kataria& Sons- New Delhi

E-Resources:

1. http://nptel.ac.in/course.php?disciplineId=115
2. http://nptel.ac.in/course.php?disciplineId=104
3. http://hperphysics.phy-astr.gsu.edu/hbase/hph.html
4. www.physicsclassroom.com
5. www.physics.org
6. www.fearofphysics.com

Course Code: 22IUCC0203B		**Course Name:** Chemistry-II
Credits: 4	**Teaching scheme:** L-3 P-0 T-1	**Evaluation Scheme:** CA–40, ESE–60
Duration of End sem Exam:3hr.		
Pre-requisites:		
Course Outcomes: At the end of the course, the students will be able to –		
CO1: Describe different processes for manufacturing metals and control of their corrosion		
CO2: Use relevant water treatment process to solve industrial problems		
CO3: Justify the selection of lubricants for different types of machines		
CO4: Describe different types of fuels and their properties and uses.		
CO5: Justify the selection of lubricants for different types of machines		
CO6: Used Relevant engineering materials in Industry		

Contents:

Unit	Content	Teaching hours
1	**Corrosion 1.1** Definition, causes of Corrosion Different types of Corrosion and factors affecting corrosion rate. **1.2 Protection of metal by**: Galvanising ,Tinning ,Electroplating ,colourizing 1.5 classification of plain carbon steel, low carbon, medium carbon,high carbon steel with their properties and Application. 1.3 **Paint:** Definition characteristics, constituent of paint and their function.	10
2	**Metallurgy** 2.1 Introduction 2.2 Definition of metallurgy Properties and application of commercial forms of iron- pig iron ,cast iron and wrought iron 2.3 Steel: Definition of steel ,Preparation of steel from pig iron using open hearth process, basic oxygen process. 2.4 classification of plain carbon steel, low carbon, medium carbon, high carbon steel with their properties and application.	10
3	**Water treatment:** 3.1 Introduction 3.2 Hardness : Classification, 3.3 Hard water in boilers and prevention: Boiler corrosion, Caustic embrittlement, priming and foaming, scales and sludge's 3.4 Water softening : lime soda process (hot lime soda and cold lime soda process) Zeolite process, Ion exchange process (Cation exchange process and anion exchange) 3.5 Potable water treatment: sedimentation, coagulation, filtration, and sterilization. Waste water treatment, BOD and COD of sewage water, Reverse Osmosis, recycling of waste water	10
4	**Fuels** 4.1 Introduction 4.2 Definition of fuel, calorific and ignition temperature. Characteristic of good fuel classification of fuels 4.3 Solid fuels: Coal classification and composition, proximate analysis, Ultimate analysis. Determination of gross calorific value by using Bomb calorimeter. 4.4 Liquid fuel: Fractional distillation of crude	10

	petroleum , boiling range, composition, properties, application of Bio- diesel 4.5 Gaseous fuel: Composition, properties ,application of Biogas, LPG, CNG	
5	**Lubricants** 5.1 Introduction Definition of lubricants .Function of lubricants, Classification of lubricants. 5.2 Characteristic : Physical characteristic of lubricants : Viscosity, Viscosity index, oiliness, flash point, fire point, 5.3 chemical characteristic of lubricants- acid value, emulsification, saponification value.(only definition) 5.4 Selection of Lubricants for road rollers, steam engine, sewing machine, concrete mixture, I.C. engine, cutting tools, gears.	10
6	**Non -metallic Engineering material: Polymers (Plastics, Rubber) :** 6.1Introduction 6.2 Plastics: Definition of plastic, polymer, polymerization, types of Polymerization with examples. Types of plastic- thermo softening plastics and thermosetting plastics and their difference, properties and applications of plastics. 6.3 Rubber: Types of rubber. Natural Rubber- definition, drawbacks of natural rubber, vulcanization of rubber with chemical reaction, applications of vulcanized rubber Synthetic rubber- definition, difference between natural and synthetic rubber, examples of synthetic rubber, properties of synthetic rubber like -elasticity, tack, and abrasion resistance, their definition and related application Thermal Insulators 6.4 Thermal Insulators -Definition, characteristics of thermal insulators, classification- organic and inorganic thermal insulators, their examples, preparation, properties and applications of polystyrene and Distilled or pure water	10

Reference Materials

1. Jain P.C & Jain Monica, Engineering Chemistry, DhanpatRai& Sons, Delhi.
2. Bhal&Tuli, Text book of Physical Chemistry, S. Chand & Company, New Delhi.
3. O. G. Palanna , Engineering Chemistry, Tata McGraw-Hill Publication, New Delhi.
4. S. S. Dara, A textbook of Engineering Chemistry, McGraw-Hill Publication, New Delhi.
5. ShashiChawla, Engineering Chemistry, DhanpatRai and Co Education and Technica Publishers
6. V. R Gowarikar, Polymer Science, New Age International Ltd.
7. B. K. Sharma, Instrumental Methods of Chemical Analysis, Krishna Prakashan Media
8. Barrow G.M., Physical Chemistry, McGraw-Hill Publication, New Delhi.
9. ShikhaAgrawal, Engineering Chemistry- Fundamentals and Applications, Cambridge Publishers.
10. Atkins, Physical Chemistry, OUP Oxford.
11. Engineering Chemistry, Wiley India Pvt, Ltd, First edition.

E-Resources:

1. NPTEL Basic Courses Engineering Chemistry (https://nptel.ac.in/courses/122/101/122101001/
2. https://www.ncertbooks.guru/engineering-chemistry/
3. Coursera Chemistry Courses (https://www.coursera.org/browse/physical-science-andengineering/chemistry?languages=en&page=2
4. “Introduction to Corrosion” NPTEL Course (https://www.youtube.com/playlist?list=PL8lylDWRkaW8BXestE4baRnN0699S11Lq
5. “Tribology &LubricationNPTEL Course (https://www.youtube.com/playlist?list=PLLwnvFqJAltJvWAfEVU4gcUK27hEA7FD)
6. “Electrochemistry” NPTEL Course (https://www.youtube.com/playlist?list=PLVFqK_9GOGXnnriQpsn0z1Rss96Rh0vsm)

Course Code: 22IUCC0204H	**Course Name:** Universal Human Values
Credits: 2**Teaching Scheme:** L-2 P-0	**Evaluation Scheme:** ESE–50
Duration of End Sem. Exam: 1 hr.	
Pre-requisites:	
Course Objectives:	

Course Outcomes: At the end of the course, the students will be able to –
CO-1 To create awareness on Engineering Ethics and Human Values.
CO-2 Understand the significance of value inputs in a classroom and start applying them in their life and profession
CO-3 Distinguish between values and skills, happiness and accumulation of physical facilities, the Self and the Body.**C**
O-4 Understand the role of a human being in ensuring harmony in society and nature.
CO-5 Distinguish between ethical and unethical practices, and start working out the strategy to actualize a harmonious environment wherever they work.

Contents –

Unit	Content	Teaching hours
1	**HUV Introduction** 1.1 Need, Basic Guidelines Content and Process for Value Education, 1.2 Continuous Happiness and Prosperity, 1.3 Self-Exploration, Continuous Happiness and Prosperity 1.4 Right Understanding, Relationship and Physical Facilities 1.5 Understanding Happiness and Prosperity Correctly.	8
2	**Harmony of Human Being** 2.1 Harmony in Myself 2.2 Understanding Human Being as a co-existence of the Sentient 'I' and the material 'Body' 2.3 Programs to ensure Sanyam and Swasthya.	8
3	**Harmony in Human-Human Relationship** 3.1 Understanding Harmony in the Family 3.2Understanding the meaning of Vishwas, 3.3Understanding the meaning of Samman, 3.4Understanding the Harmony in the Society, 3.5Visualizing a Universal Harmonious order in Society.	8
4	**Harmony in the Nature and Existence** 4.1 Understanding the Harmony in the Nature 4.2 Interconnectedness and Mutual Fulfilment among the Four Orders of Nature 4.3 Understanding Existence as Co-existence.	8
5	**Harmony on Professional Ethics** 5.1 Natural Acceptance of Human Values, 5.2 Definitiveness of Ethical Human Conduct, 5.3 Basis for Humanistic Education, 5.4 Humanistic Constitution and Humanistic Universal Order, 5.5 Competence in Professional Ethics, Case Studies of Typical Holistic Technologies.	8
6	**SELF DEVELOPMENT**	8

	6.1 Goal setting and its importance. 6.2 Characteristics of Goal setting (SMART-Specific, Measurable, Attainable, Realistic, Time bound) 6.3 Time Management - Importance, prioritization of work, time matrix, time savers, and time wasters. 6.4 Stress Management - Definition, types of stress, causes of stress, managing stress, and stress busters.	

Reference Material:
1 Corliss Lamont, Philosophy of Humanism 2.Gaur. R.R. ,Sangal. R, Bagaria. G.P, A Foundation Course in Value Education, Excel Books, 2009. 3.Gaur. R.R. ,Sangal. R ,Bagaria. G.P, Teachers Manual Excel Books, 2009. 4.I.C. Sharma . Ethical Philosophy of India Nagin& co Julundhar 5.Mortimer. J. Adler, – Whatman has made of man 6.William Lilly Introduction to Ethic Allied Publisher
Online Resources: 1. https://www.britishcouncil.in/english/learn-online 2. http://learnenglish.britishcouncil.org/en/content 3. http://www.talkenglish.com 4. Languagelabsystem.com 5. www.wordsworthelt.com

Course Code: 22IUCC0205T	**Course Name:**Computer Workshop
Credits: 2**Teaching Scheme:** L-2 P-0 T-0	**Evaluation Scheme:** Audit
Duration of End Sem Exam:	
Pre-requisites:	
Course Objectives:	
Course Outcomes: At the end of the course, the students will be able to –	
CO1: Troubleshoot common motherboard problems.	
CO2: Select processors required for relevant systems.	
CO3: Partition/Format hard disk drives.	
CO4: Troubleshot peripherals and networks.	
CO5: Test Power supplies	
CO6: Networking and Maintenance	

Contents:

Unit	Content	Teaching hours
1	**Motherboard** Motherboard: Components Layout Connections Motherboard Types: Types and features Enhancing features of motherboard: Adding and or replacing components. Troubleshooting: problems of motherboard.	4
2	**Processor and BIOS** Processor: Common features, Types, Basic structure of CPU,Different levels of cache, system bus, clock speed, packaging. Multiple core processors: Description, Two core processor architecture and multicourse architecture. Co- Processors: Graphics, math. BIOS: Basic I/O system services, Bios interaction, date and time, Boot device priority, Boot setting configuration, password security.	4
3	**Hard Disk** Hard disk Interfaces:EIDE,Serial ATA,SCSI,USB and IEEE 1394,RAID,Solid state Drive. Disk Structures: Heads, tracks, sectors, cylinders, clusters, landingzone, MBR,Zone bit recording. Disk Performance parameters characteristics: Seeks and latency, Data transfer rate. File system: FAT16,FAT32,NTFS,Unix file system,EXT2/EXT3,RAID	4
4	**I/O and Modem** Troubleshoot I/O devices: Keyboard, Switches, Mouse, Scanners, webcam,monitors,printers,speaker and Mike,LCD projector. I/O cables: Specifications of I/O cables,Types of I/O cables,Types of I/O ports,Internal and external modem, Block diagram and specification. Network Interfaces: Definition of network interface, Types of network interface, troubleshooting of network connectivity, Antivirus.	4

5	**Power supply** Purpose and features of SMPS,Working of SMPS Fault finding in power supply Uninterrupted Power supply: Characteristics of UPS,Types of UPS,online and offline. Preventive Maintenance of power supply.	4
6	**Networking and Maintenance** Share and control resource and Internet connection through network. Implement Network Security to protect from various attacks on networking.MAC Address and its features. Firewall : Technology, implementaion	6

Reference Material:

1. James. K. L. "The Computer hardware installation, interfacing, troubleshooting and maintenance", PHI Learning ,New Delhi,2014 ISBN:978-81-203-4798-4

2. Gupta, Vikas ,"Comdex: Hardware and Networking course kit" Dreamtech Press, New Delhi ISBN:978-93-5119-265-7

3. Minasi,Mark, "The Complete Pc Upgrade and maintenance Guide" BPB Publication, Mumbai, New Delhi ISBN:978-81-265-0627

Online Resources:

1. https://www.tutorialspoint.com/computer_fundamentals/computer_motherboard.htm
2. https://www.javatpoint.com/what-is-a-multicore-processor
3. https://www.geeksforgeeks.org/co-processor-computer-architecture/
4. https://www.digitalocean.com/community/tutorials/an-introduction-to-networking terminology-interfaces-and-protocols
5. https://www.tutorialspoint.com/information_security_cyber_law/network_security.htm
6. https://www.tutorialspoint.com/what-is-a-proxy-server

Course Code: 22IUCC0202L	**Course Name:** Physics-II Lab
Credits: 1 **Teaching scheme:** L-0 P-2 T-0	**Evaluation scheme:** TW–50
Pre-requisites:	

Course Outcomes: At the end of the course, the students will be able to –
CO 1: To learn about the useful solid material in industry & understand its physical properties.
CO 2: To learn about the useful liquid material & understand its physical properties.
CO 3: Understand the concepts of sound wave
CO 4: To learn about the concepts of radioactive and ultrasonic principles.
CO 5: To learn about the concepts of electrostatic & X rays
CO 6: To learn about the concepts of the principles of electricity , magnetism & semiconductors to solve engineering problems.

Contents:

Unit	Content	Teaching hours
1	Use Searle's method to determine the Young's modulus of given wire	2
2	Determine the coefficient of viscosity of given liquid by Stoker's materials	2
3	Find the downward force, along an inclined plane, acting on a roller due to gravity and its relationship with the angle of inclination.	2
4	Determine the refractive index of glass slab	2
5	Determine the refractive index of prism using Pin Method	2
6	Use resonance tube to determine velocity of sound.	2
7	Determine the specific resistance of given wire.	2
8	Use magnetic compass to draw the magnetic lines of forces of magnet of different shapes.	2
9	Use p-n junction diode to draw forward bias and reverse bias I-V characteristics.	2

Course Code:22IUCC0203L	**Course Name:** Chemistry-II Lab
Credits: 1 Teaching scheme: L-0 P-2 T-0	**Evaluation scheme: TW–50**
Pre-requisites:	
Course Outcomes: At the end of the course, the students will be able to	
CO1: Describe different processes for manufacturing metals and control of their corrosion **CO2:** Use relevant water treatment process to solve industrial problems **CO3:** Justify the selection of lubricants for different types of machines **CO4:** Describe different types of fuels and their properties and uses. **CO5:** Used Relevant engineering materials in Industry	

Contents –

Unit	Content	Teaching hours
1	Prepare the corrosive acidic medium for Aluminum	2
2	Determine the rate of corrosion on different temperatures for Aluminum	2
3	Determine the acid base titration neutralization reaction	2
4	Determine the percentage of iron present in given Hematite ore by KMnO4 solution	2
5	Determine the percentage of copper in given copper ore	2
6	Determine Chloride content in given water sample by Mohr's method.	2
7	Determine total hardness, temporary hardness and permanent hardness of watersample by EDTA method	2
8	Determine the alkalinity of given water sample	2
9	Determine the effect of temperature on viscosity for given lubricating oil using Redwood Viscometer	2
10	Determine the moisture and ash content in given coal sample by using proximate analysis	2
11	Prepare phenol formaldehyde resin used in manufacturing of Bakelite plastic	2
12	Prepare Urea formaldehyde resin and understand the structure and properties for its applications in engineering	2

Course code: 22IUCC0205L	**Course name: Computer Workshop Lab**
Course category: PCC	
Credits: 01 **Teaching scheme:** P-2Hrs/Week **Evaluation scheme:** TW–50, PR–00	
Pre-requisites: 1	
Laboratory Outcomes: At the end of the session, the students will be able to -	
LO1: Troubleshoot common motherboard problems.	
LO2: Select processors required for relevant systems.	
LO3: Partition/Format hard disk drives.	
LO4: Troubleshot peripherals and networks.	
LO5: Test Power supplies	
LO6: Networking and Maintenance	

List of Experiments–

Sr.No.	Name of Experiment
1	Identify different hardware components onmotherboard.
2	Troubleshoot common problems of motherboard.
3	Configure BIOS setting.
4	Partition and manage hard disk: Format harddrives with different file systems
5	Install operating system –Windows family
6	Install operating system –Unix family
7	Troubleshoot hard disk problems
8	Set Keyboard, Mouse, Monitor, Speaker, Webcam, Microphone and LCD Projector ect.
9	Install SMPS,measure voltage levels in mainconnectors of SMPS connecting various subsystems.
10	Troubleshoot computer system by diagnosing the problem.
11	Use diagnostic software for fault finding andviruses
12	Undertake preventive maintenance of PC using vacuum cleaner and simple tools.
13	Configure internet connection to the PC usingwireless technology and troubleshoot various connection related problems
14	Install Proxy Server and configure it.
15	Set up basic protection using public keys and MAC address filters.
16	Practice on firewall technologies to secure the network perimeter

Reference Books:

1. B Ram –Computer Fundamental architecture and Organization, New age international Publications
2. DrRajendraKawale ,"Computer Fundamental", Devraj Publications Dist. Solapur

Online Resources: 1. https://www.tutorialspoint.com

Course Code: 22IUCC0204L	**Course Name:** Universal Human Values Lab
Credits: 1 Teaching scheme: P-2	**Evaluation scheme: TW–50**
Course Objectives:	
Course Outcomes: At the end of the course, the students will be able to –	
CO-1 To create awareness on Engineering Ethics and Human Values.	
CO-2 Understand the significance of value inputs in a classroom	
and start applying them in their life and profession	
CO-3 Distinguish between values and skills, happiness and accumulation of physical facilities, the Self and the Body.	
CO-4 Understand the role of a human being in ensuring harmony in society and nature.	
CO-5 Distinguish between ethical and unethical practices, and start working out the strategy to actualize a harmonious environment wherever they work.	

Contents –

Unit	Content	Teaching hours
1	Introduce yourself in detail	2
2	Explain your ideas of success/progress/development.	2
3	Write down 5 needs of your self and 5 needs of your body	2
4	Explain self- esteem with two examples	2
5	Chalk out programs to ensure that you are responsible to your body- for the nurturing, protection and right utilization of the body.	2
6	Form small groups in the class and in that group initiate dialogue and ask the eight questions related to trust	2
7	Write a note in the form of story, poem, skit, essay, narration, dialogue to educate a child. Evaluate it in a group.	2
8	Choose any one subject being taught today. Evaluate it and suggest suitable modifications to make it appropriate and holistic.	2
9	Explain harmony between body and mind in your own words.	2
10	Suggest some ideas to be a happy human being.	2
11	Collect information on any two great humanitarians.	2
12	Explain importance of goal setting .	2

Course Code:22IUCC0206L	**Course Name:**Workshop practice Lab
Credits: 2 Teaching scheme: P-4	**Evaluation scheme: TW–100**
Pre-requisites: Basic knowledge of computer, Computer Aided Drawing I lab	
Course Outcomes: At the end of the course, the students will be able to –	
CO 1: Select tool and machinery according to job.	
CO 2: Use hand tools in different shops for performing different operation.	
CO 3: Opera equipment and machinery in different shop.	
CO 4: Prepare composite or utility job according to drawing.	
CO5: Maintain workshop tools and equipment's	

Contents –

Unit	Content	Teaching hours
1	Prepare fitting job (male and female type) as per given drawing or job with following operation marking operation punching operation drilling operation cutting operation filling operation fitting operation (male and female assembly) checking correctness of fit of mating parts	8
2	Prepare bill of material along with estimate cost according given drawing of job such as repairing of class room furniture / book shelves / metallic doors / motor sari guard / battery stand with locking device considering the following applicable operation. a) marking operation as per drawing . b)cutting operation as per drawing . c) cleaning operation as per drawing d) edge preparation operation as per drawing e) filling operation as per drawing f) bending operation as per drawing g) welding operation as per drawing	8
3	Prepare steel frame / structure of utility job (like stool, benches, tables, drawing desk, window grill, ventilator, door frame or similar job) involving arc welding joints as per given drawing or job.	8
4	Prepare simple turning job with facing/taper turning/knurling/grooving/threading operation as per drawing	8
5	Prepare sheet metal utility job using following operation - Part I a. Cutting and Bending b. Edging c. End Curling d. Lancing e. Soldering f. Riveting	8